



III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
C	B	C	B	A	C	A	B	A	B	C	C	A	A	A

16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
C	D	D	C	B										

- 1 C The rate of collision f of a molecule with a wall is given by

$$f \propto v/2L$$

where L is the dimensions of the box and v is the speed of the molecules.

The rate of collision increases with

- increasing the temperature or internal energy (which in turn increases the speeds of all molecules)
- decreasing the dimensions of the box (Hence C is not true)

Although the speeds of the gas molecules are different, the pressure of the gas depends on the mean square speed which is constant if the temperature is constant.

Note: Proportionality is used here because the motion of these gas molecules may not be normal to the surface.

- 2 B Centigrade scale depends on the thermometric properties of the thermometer while the thermodynamic temperature scale is independent of thermometric properties of any materials or thermometers.

- 3 C From $p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle = \frac{1}{3} \rho \langle c^2 \rangle$,

$$\sqrt{\langle c^2 \rangle} = \sqrt{\frac{3p}{\rho}} = \sqrt{\frac{3 \times 100 \times 10^3}{0.180}} = 1290 \text{ m s}^{-1}$$

- 4 B Using $pV = nRT$,

$$p \propto T$$

$$p = p_{atm} + p_{gauge}$$

Pressure in tyre at 65°C

$$p_2 = \frac{T_2}{T_1} \cdot p_1 = \frac{65 + 273.15}{28 + 273.15} \times [(160 + 100) \times 10^3] = 292 \text{ kPa}$$

So final pressure gauge reading is 192 kPa. The percentage increase in pressure gauge reading

$$\frac{\Delta p_{gauge}}{p_{gauge}} \times 100\% = \frac{192 - 160}{160} \times 100\% = 20\%$$

- 5 A The change in momentum Δp of a molecule colliding normally with a wall is $|\Delta p| = 2mu$ where m and u are the mass and speed of the molecule.

- 6 C Average K.E. of molecules

$$\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT \Rightarrow \sqrt{\langle c^2 \rangle} \propto \sqrt{\frac{1}{m}}$$

if thermodynamics temperature T is constant.

So r.m.s. speed of hydrogen molecules

$$\sqrt{\langle c_H^2 \rangle} = \sqrt{\frac{m_{Ne}}{m_H}} \times \sqrt{\langle c_{Ne}^2 \rangle} = \sqrt{\frac{20u}{2u}} \times 400 = 1300 \text{ m s}^{-1}$$

- 7 A Since equilibrium has been established, the pressure p in X and Y are equal. Using $pV = nRT$,

$$p = \frac{n_A RT_A}{V_A} = \frac{n_B RT_B}{V_B}$$

Rearranging,

$$\frac{n_B}{n_A} = \frac{T_A}{T_B} \times \frac{V_B}{V_A} = \frac{200}{400} \times \frac{1}{2} = \frac{1}{4}$$

$$n_B = \frac{1}{4} n_A = \frac{1}{4} b$$

- 8 B Using $pV = nRT$,

$$n_x = \frac{p_x V_x}{RT_x} = \frac{3PV}{2RT}$$

$$n_y = \frac{p_y V_y}{RT_y} = \frac{PV}{4RT}$$

The mass of gas in connected vessels is conserved, and pressure P_f in X and Y are equal at equilibrium.

$$n_f = n_x + n_y$$

$$\Rightarrow \frac{P_f V_f}{RT_f} = \frac{3PV}{2RT} + \frac{PV}{4RT}$$

Since $V_f = \frac{5}{2}V$ and $T_f = T$,

$$P_f = \frac{2RT}{5V} \left(\frac{7PV}{4RT} \right) = 0.70P$$

- 9 A Average K.E. of molecules

$$\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT \Rightarrow \langle c^2 \rangle \propto T$$

$$\text{So } T_2 = \frac{\langle c_2^2 \rangle}{\langle c_1^2 \rangle} \times T_1 = (0.8)^2 \times 295 = 188.8 \text{ K} = -84.4^\circ \text{C}$$

$$\text{Note: } \sqrt{\langle c_2^2 \rangle} = 0.8\sqrt{\langle c_1^2 \rangle}.$$

- 10 B Using $pV = nRT$,

$$n \propto \frac{V}{T}$$

since the pressure p in X and Y are equal. So

$$\frac{n_X}{n_Y} = \frac{V_X}{V_Y} \cdot \frac{T_Y}{T_X} = 2 \times \frac{1}{2} = 1$$

- 11 C Average K.E. of molecules

$$\langle \text{K.E.} \rangle = \frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT \Rightarrow \langle \text{K.E.} \rangle \propto \langle c^2 \rangle \propto T$$

Since gas molecules are moving twice as fast, r.m.s. speed is doubled (mean squared speed is quadrupled),

$$\langle \text{K.E.}_2 \rangle = \frac{\langle c_2^2 \rangle}{\langle c_1^2 \rangle} \times \langle \text{K.E.}_1 \rangle = 4 \times \langle \text{K.E.}_1 \rangle = 4 \times 1.6 \times 10^{-23} = 6.4 \times 10^{-23} \text{ J}$$

$$T_2 = \frac{\langle c_2^2 \rangle}{\langle c_1^2 \rangle} \times T_1 = 4T_1 = 4 \times 573 = 2292 \text{ K}$$

- 12 C Using $pV = nRT$, the amount of gas that escaped

$$n_{\text{escaped}} = n_i - n_f = \frac{p_i V_i}{RT_i} - \frac{p_f V_f}{RT_f} = \frac{3.0 \times 10^5 \times 1.0}{8.31 \times 200} - \frac{3.0 \times 10^5 \times 1.0}{8.31 \times 400} = 90.25 \text{ mol}$$

Volume of escaped gas at atmospheric pressure & room temperature

$$V_{\text{escaped}} = \frac{n_{\text{escaped}} RT}{p} = \frac{90.25 \times 8.31 \times 300}{1.0 \times 10^5} = 2.25 \text{ m}^3$$

- 13 A Average K.E. of molecules

$$\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$$

$$\Rightarrow \sqrt{\langle c^2 \rangle} \propto \sqrt{T}$$

$$\text{So } \langle c_2 \rangle = \sqrt{\frac{T_2}{T_1}} \times \langle c_1 \rangle = \sqrt{2}c$$

- 14 A Using $pV = nRT$, the amount of gas pumped into container

$$\Delta n = n_f - n_i = \frac{\Delta p V}{RT} = \frac{(620 - 500) \times 10^3 \times 4.0 \times 10^{-4}}{8.31 \times 300} = 0.019 \text{ mol}$$

- 15 A Using $pV = nRT$,

$$n \propto V \propto L$$

since the gases in both chambers are at the same pressure and temperature.

Therefore

$$\frac{L_1}{L_2} = \frac{n_1}{n_2} = \frac{20 \times 10^{-3} / 0.028}{45 \times 10^{-3} / 0.004} = 0.063$$

- 16 C $\langle E_k \rangle = \frac{3}{2} kT$ which is independent of the mass of the molecules, i.e., molecules at the same temperature have the same average K.E.

- 17 D From the ideal gas equation $pV = nRT$,

$$p = \frac{nR}{V} \times T = \frac{nR}{V} \theta + \frac{nR}{V} (273.15)$$

So in a graph of p against θ , $\frac{nR}{V}$ is the gradient and $\frac{nR}{V} (273.15)$ is the vertical intercept.

Since $V_x > V_y$, the gradient and vertical intercept for vessel X is less than of vessel Y.

- 18 D From $p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle = \frac{1}{3} \rho \langle c^2 \rangle$,

$$\sqrt{\langle c^2 \rangle} = \sqrt{\frac{3p}{\rho}} = \sqrt{\frac{3 \times 150 \times 10^3}{0.178}} = 1600 \text{ m s}^{-1}$$

- 19 C Using $pV = NkT$,

$$N = \frac{pV}{kT} = \frac{2.5 \times 10^5 \times 3.0 \times 10^{-4}}{1.38 \times 10^{-23} \times (27 + 273.15)} = 1.8 \times 10^{22}$$

- 20 B The change of momentum Δp of the molecules on collision with the wall is not negligible.
For a molecule colliding normally with a wall,

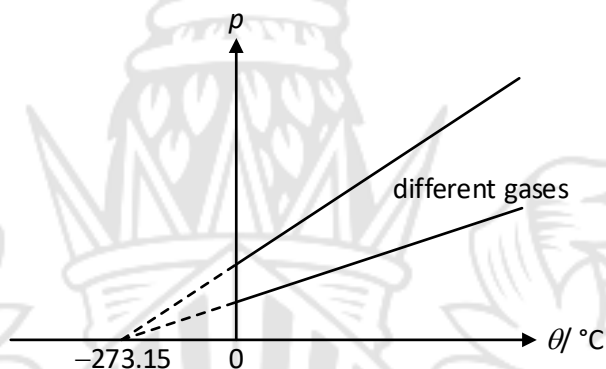
$$|\Delta p| = 2mu$$

where m and u are the mass and speed of the molecule.

Structured Questions

1. [AJC/H2/Prelims 2018/P2/3]

- a. i. Absolute zero is the theoretical temperature at which the molecules of all substances have the lowest/minimum kinetic energy.
- ii. Data from experiments using the constant-volume gas thermometers are collected and plotted on a graph of pressure against temperature.
- By extrapolating the graph to zero pressure, the temperature tends to a value of -273.15°C .



When the experiment is repeated using different gases, this temperature remains the same.

This temperature is known as the absolute zero and it does not depend on the property of any particular substance.

- b. Using ideal gas equation, $pV = nRT$

Initially for cylinder B

$$n_B = \frac{p_B V_B}{RT_B} = \frac{1.2 \times 10^5 \times 2.0 \times 10^{-2}}{8.31 \times (273.15 + 37)} = 0.93 \text{ mol}$$

When tap is opened, for both cylinders

$$p_f = \frac{(n_A + n_B)RT_f}{V_A + V_B}$$

For the system consisting of two cylinders, $\Delta U = 0$ since both cylinders are insulated, i.e., $Q = 0$, and total volume is fixed, i.e., $W = 0$.

$$T_f = T_i = 37^\circ\text{C}$$

$$p_f = \frac{(1.2 + 0.93) \times 8.31 \times (273.15 + 37)}{2 \times 2.0 \times 10^{-2}} = 1.4 \times 10^5 \text{ Pa}$$

2. [IJC/H2/Prelims 2018/P3/3]**a. Any 2 assumptions**

- There are sufficiently large number of molecules and they move about in random motion.
- The volume of molecules is negligible compared with the volume occupied by the gas.
- The molecules exert no intermolecular forces on one another except during collisions.
- The time during a collision is negligible compared with the time between collisions.

b. i. Mass of one atom of He-4

$$m = \frac{M_r}{N_A} = \frac{4.0 \times 10^{-3}}{6.02 \times 10^{23}} = 6.6 \times 10^{-27} \text{ kg}$$

ii. Internal Energy

$$U = \frac{3}{2} nRT = \frac{3}{2} \times 1.2 \times 8.31 \times (273.15 + 27) = 4.5 \times 10^3 \text{ J}$$

Alternatively, one can use $U = \frac{3}{2} NkT = \frac{3}{2} nN_A kT$.

iii. Average K.E. of molecules

$$\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT \Rightarrow \sqrt{\langle c^2 \rangle} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3 \times 1.38 \times 10^{-23} \times (273.15 + 27)}{6.6 \times 10^{-27}}} = 1.4 \times 10^3 \text{ m s}^{-1}$$

- iv.** At any temperature, the helium atoms move with a range of speeds that can be faster or slower than r.m.s. speed. A small fraction of these atoms will have speeds that exceeds the escape speed of Earth and hence are fast enough to escape the Earth's atmosphere.

3. [JJC/H2/Prelims 2018/P3/8]**a. i. Any 2 assumptions**

- There are sufficiently large number of molecules and they move about in random motion.
- The molecules exert no intermolecular forces on one another except during collisions.
- Collisions between molecules and the container are perfectly elastic.
- The time of collision is negligible compared with the time between collisions.
- The volume of molecules is negligible compared with the volume occupied by the gas.

ii. 1. Average kinetic energy of He $\langle E_k \rangle = \frac{3}{2} kT = \frac{3}{2} \times 1.38 \times 10^{-23} \times 290 = 6.00 \times 10^{-21} \text{ J}$

- 2.** Since mean square speed of molecules is inversely proportional to the mass of the molecules, the mean square speed of nitrogen molecules will different from that of helium molecules as their molecular masses are different.

4. [MI/H2/Prelims 2018/P3/3 Part]**a. i. Since the motion of gas molecules are random, the gas molecules possess no preferred direction of motion and hence the average velocity of the gas molecules is zero.**

Since momentum (a vector) is the product of mass and velocity. Average momentum of the gas molecules of same mass is also zero.

ii. When the gas molecules hit the wall of the container, they rebound off the wall with no change in speed (elastic collision). But these molecules experience a change in momentum (change in direction). According to N2L, the change in momentum of these molecules is due to the force (rate of change of momentum of all molecules) exerted by the walls of the container.

By Newton's third law, these molecules exert a force, equal in magnitude but opposite in direction, on the walls. Due to the large number of molecules moving in random directions and speeds, we can derive the average force per unit area on the wall. This is the gas pressure on the walls.

5. [NYJC/H2/Prelims 2018/P3/5 Part]

- a. The internal energy of a system is the sum of the kinetic energy due to the random motion of the molecules, and the potential energy, associated with the intermolecular forces of the system.
- b. ρ is the density of the gas. $\langle c^2 \rangle$ is the mean-square speed of the atoms/molecules in the gas.
- c. According to kinetic theory of gases,

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle \Rightarrow pV = \frac{1}{3} Nm \langle c^2 \rangle$$

where N is the number of molecules in the gas; m the mass of each molecule and V the volume of the gas.

Total kinetic energy

$$\text{K.E.}_{\text{total}} = \frac{1}{2} Nm \langle c^2 \rangle = \frac{3}{2} \times \frac{1}{3} Nm \langle c^2 \rangle = \frac{3}{2} pV$$

6. [PJC/H2/Prelims 2018/P3/3]

- a Any 2 assumptions
- There are sufficiently large number of molecules and they move about in random motion.
 - The molecules exert no intermolecular forces on one another except during collisions.
 - Collisions between molecules and the container are perfectly elastic.
 - The time of collision is negligible compared with the time between collisions.
 - The volume of molecules is negligible compared with the volume occupied by the gas.

- b. i. According to kinetic theory of gases,

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle \Rightarrow pV = \frac{1}{3} \rho \langle c^2 \rangle$$

$$\sqrt{\langle c^2 \rangle} = \sqrt{\frac{3p}{\rho}} = \sqrt{\frac{3 \times 1.8 \times 10^5}{2.7}} = 450 \text{ m s}^{-1}$$

- ii. 1. Internal energy of an ideal monoatomic gas

$$U = \frac{3}{2} nRT \Rightarrow \Delta U = \frac{3}{2} nR\Delta T = \frac{3}{2} \times 2.5 \times 8.31 \times (576 - 288) = 8.97 \times 10^3 \text{ J}$$

2. Work done on gas $W = -p\Delta V = -nR\Delta T = -2.5 \times 8.31 \times (576 - 288) = -5.98 \times 10^3 \text{ J}$

3. 1st law of thermodynamics

$$\Delta U = Q + W \Rightarrow Q = \Delta U - W = 8.97 \times 10^3 - (-5.98 \times 10^3) = 1.50 \times 10^4 \text{ J}$$

7. [YJC/H2/Prelims 2018/P2/3]

- a. i. According to kinetic theory of gases,

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle \Rightarrow pV = \frac{1}{3} \rho \langle c^2 \rangle$$

$$\sqrt{\langle c^2 \rangle} = \sqrt{\frac{3p}{\rho}} = \sqrt{\frac{3 \times 2.1 \times 10^5}{1.5}} = 650 \text{ m s}^{-1}$$

- ii. Average K.E. of molecules

$$\langle \text{K.E.} \rangle = \frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT \Rightarrow \frac{M_r}{N_A} \langle c^2 \rangle = 3kT$$

since mass of each molecule $m = M_r / N_A$.

Temperature of nitrogen gas

$$T = \frac{M_r \langle c^2 \rangle}{3kN_A} = \frac{28.0 \times 10^{-3} \times 650^2}{3 \times 6.02 \times 10^{23} \times 1.38 \times 10^{-23}} = 475 \text{ K}$$

Note: Molar mass of nitrogen gas should be 28.0 g.

- iii. The root-mean-square (r.m.s.) speed of the molecules increases with increasing temperature of the gas. This results in greater change in momentum for each collision with the wall and the higher frequency of collisions of these molecules with the walls.

According to N2L, a greater average force (rate of change of momentum of all molecules) acts on the molecules during collisions with any wall. By N3L, these molecules exert a greater average force and pressure (average force per unit area) on the walls.

